

Figure Notebooks — Equation Reference

AD-presynaptic-calcium-model / figures/

1. Single-AP kinetics & buffering

fig3_single_AP, fig4_PSblocked_paired_pulse, figS1_model_fits_timescales

AP upstroke (true stimulus time)

Used in: fig3, figS1

$$t_{\text{stim}} = t [\operatorname{argmax}_t (dV/dt)]$$

dV/dt via a numerical gradient of the AP.dat voltage trace against time.

Calbindin-bound Ca^{2+}

Used in: fig3, fig4, figS1

Total Ca^{2+} bound across calbindin's 8 modeled binding states, baseline-subtracted at the stimulus time and zeroed before it.

$$[\text{Ca}]_{\text{calbindin}}(t) = (1/100) \cdot \sum_{i=1}^8 \text{calB}_i(t) - \text{baseline}$$

SERCA-bound Ca^{2+}

Used in: fig3, fig4, figS1

Total Ca^{2+} bound across SERCA's singly- and doubly-bound states (cytosol- and lumen-facing), doubly-bound states counted twice for their two bound ions.

$$[\text{Ca}]_{\text{SERCA}}(t) = (1/100) \cdot [X1(t) + 2 \cdot X2(t) + 2 \cdot Y2(t) + Y1(t)] - \text{baseline}$$

PMCA net extrusion

Used in: fig3, fig4

$$J_{\text{PMCA,net}}(t) = (1/100) \cdot [\text{flux}(t) + \text{leak}(t)] - \text{baseline}$$

2. Curve-fitting & timescale toolkit

figS1_model_fits_timescales

Rise time (10-90%)

Used in: figS1

Model-free rise-time metric: time between the (linearly interpolated) crossings of 10% and 90% of the way from baseline to peak.

$$t_{\text{rise}} = t(0.9) - t(0.1), \quad t(f) = \text{time where } y \text{ crosses baseline} + f \cdot (\text{peak} - \text{baseline})$$

Decay time to 1/e

Used in: figS1

Model-free decay-time metric.

$$t_{\text{decay}} = t_{\text{cross}} - t_{\text{peak}}, \quad t_{\text{cross}} = \text{first post-peak time where } y \leq \text{baseline} + (1/e) \cdot (\text{peak} - \text{baseline})$$

$1/e \approx 0.3679$.

Exponential relaxation fits

Used in: figS1

Mono- and bi-exponential forms fit via nonlinear least squares (optionally weighted by $1/|y|$). Goodness of fit reported throughout as R^2 .

$$\text{decay: } y(t) = A \cdot e^{-t/\tau} + C$$

$$\text{rise: } y(t) = A \cdot (1 - e^{-t/\tau}) + C$$

$$\text{bi-exp decay: } y(t) = A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2} + C \quad (\tau_1 < \tau_2, \text{ fast/slow})$$

$$\text{bi-exp depletion: } y(t) = C - A_1(1 - e^{-t/\tau_1}) - A_2(1 - e^{-t/\tau_2})$$

$$R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$$

Sigmoid (logistic) fit for driven responses

Used in: figS1

Standard voltage-gated-channel-activation functional form, used for responses driven directly by the stimulus rather than relaxing passively.

$$y(t) = C + A / (1 + e^{-(t-t_{\text{half}})/\tau})$$

Adaptive-shift fitting

Used in: figS1

Refits the relevant decay/rise model at increasing onset offsets (0, 1, 2, ... up to 25 samples past a reference index) until $R^2 \geq 0.995$ is reached, or the best R^2 found otherwise. Anchoring an exponential fit exactly at a true extremum biases τ , since a sum of exponentials is maximally steep at its own $t = 0$.

PMCA joint onset + steady-extrusion model

Used in: figS1 only

Fits an onset saturating exponential and a constant-rate linear extrusion term simultaneously, separating PMCA's activation time constant from its steady-state extrusion rate ($R^2 > 0.999$).

$$J_{\text{PMCA}}(t) = A \cdot (1 - e^{-t/\tau}) + \text{rate} \cdot t + C$$

Akaike Information Criterion (model comparison)

Used in: figS1 — point-process release-timing models (§4.3) and single- vs. bi-exponential ER-depletion fits

$$\text{AIC} = 2k - 2 \cdot \ln L$$

k = number of free parameters, L = maximized likelihood; the preferred model is $\text{argmin}(\text{AIC})$.

3. RyR / ER flux & depletion

fig6_RyR_train (canonical); fig7_ER_hyperactivity, figS3_ER_hyperactivity_RyRunder250

Net RyR Ca^{2+} flux

Used in: fig6, fig7, figS3

$$J_{\text{net}}(t) = J_{\text{out}}(t) - J_{\text{in}}(t)$$

Pulse-windowed peak/baseline extraction

Used in: fig6, fig7, figS3

Splits a train trace into one window per pulse to extract a per-pulse peak and pre-pulse baseline.

$$\text{window } k = [t_1 + k/f, \quad t_1 + (k+1)/f), \quad \text{peak}_k = \max(\text{trace in window } k)$$

t_1 = first-pulse time, f = train frequency; baseline_k = trace value at the sample immediately preceding window k 's onset.

Steepest single-pulse ER depletion rate

Used in: fig6, fig7 (both VDCC datasets), figS3

The single largest pulse-to-pulse drop in peak ER Ca^{2+} , converted to a rate.

$$\text{rate} = \max_k [\text{ER}_{\text{peak},k-1} - \text{ER}_{\text{peak},k}] / \text{ISI}$$

ISI = 1/frequency (e.g. 50 ms at 20 Hz).

RyR flux time-to-half-max

Used in: fig6, fig7 (both VDCC datasets), figS3

$$t_{1/2} = \text{first } t \text{ where } J_{\text{net}}(t) \geq J_{\text{net}}(\infty)/2$$

$J_{\text{net}}(\infty)$ = the trace's final/asymptotic value.

4. Facilitation & vesicle-release timing models

fig3_single_AP, fig4_PSBlocked_paired_pulse, fig5_RyR_paired_pulse_facilitation, figS1_model_fits_timescales

Saturating-recovery facilitation model

Used in: fig4, fig5

Fits facilitation as a function of first-pulse release probability.

$$F(x) = [1 - (1-x)^{a \cdot x^b}] / x$$

$x = \text{Pr}(\text{pulse } 1)$; a, b fit by nonlinear least squares. (Inner exponent written as a caret, " x^b ", since it is itself an exponent of x .)

Exponential ISI-decay fit

Used in: fig5

Facilitation as a function of inter-stimulus interval (log-spaced ISI sweep, per condition); also used in place of the model above to fit $\text{Pr}(\text{pulse } 2)$ directly where it gives a better fit.

$$F(\Delta t) = A + B \cdot e^{-\Delta t/\tau}$$

Point-process release-timing models, compared by AIC

Used in: figS1

Synchronous and asynchronous release-event populations are each fit with four candidate timing models; the preferred model per population is chosen by AIC (§2.8). x_i = event times relative to stimulus onset; T = observation window length.

$$\text{uniform}(0,T): k=0, \ln L = -n \cdot \ln T$$

$$\text{delayed uniform}(d,T): k=1, d = \min(x_i) \text{ (closed form), } \ln L = -n \cdot \ln(T-d)$$

$$\text{truncated exponential}(\tau): k=1, \ln L(\tau) = -\sum x_i/\tau - n \cdot \ln \tau - n \cdot \ln(1 - e^{-T/\tau})$$

$$\text{delayed truncated exponential}(d,\tau): k=2, \ln L(d,\tau) = -\sum (x_i - d)/\tau - n \cdot \ln \tau - n \cdot \ln(1 - e^{-(T-d)/\tau}), \\ 0 \leq d < \min(x_i)$$

fig3 fits only the model family it takes as correct for each population (no AIC table shown there); figS1 shows the full four-way comparison.

5. Postsynaptic EPSC metrics

fig8_poirazi_synapse_validation, fig9_postsynaptic_train_EPSC

EPSC amplitude, 20-80% rise, 1/e decay

Used in: fig8

Applied to the dendritic and somatic current traces, and to a series-resistance sweep.

$$\text{baseline} = \text{mean}[y(t)] \text{ over } [t_{\text{ev}} - 5\text{ms}, t_{\text{ev}})$$

$$\text{amp} = \min[y(t) - \text{baseline}] \text{ (peak inward current)}$$

$$t_{\text{rise}} = t(0.8 \cdot \text{amp}) - t(0.2 \cdot \text{amp}) \text{ (post-peak segment, first crossings)}$$

$$t_{\text{decay}} = \text{first } t \text{ (post-peak) where } y \geq 0.3679 \cdot \text{amp}, \text{ minus peak time}$$

Conductance target (Ohm's law)

Used in: fig8

Converts the target experimental EPSC amplitude to a unitary conductance.

$$g = I / |V_{\text{hold}}| \quad (26.8 \text{ pA} / 70 \text{ mV} \approx 383 \text{ pS})$$

Unitary-EPSC isolation rule

Used in: fig9

A pulse counts as an isolated single-vesicle event only if exactly one vesicle was released on that pulse and none on either neighboring pulse, avoiding EPSC-summation contamination. Used as a cross-condition consistency check (flagged if the spread exceeds 1%).

$$\text{pulse } k \text{ of trial } i \text{ is isolated iff } V_{i,k}=1, V_{i,k-1}=0, V_{i,k+1}=0$$

$$\text{unitary EPSC} = \text{mean}(\text{amplitude over isolated events})$$

Amplitude, potency, failure rate

Used in: fig9

Per-pulse, per-condition summary statistics across trials.

$$\text{Amplitude}_k = \text{mean}_i(A_{i,k}) \text{ (includes failures)}$$

$$\text{Potency}_k = \text{mean}_i(A_{i,k} \mid V_{i,k} > 0) \text{ (successes only)}$$

$$\text{Failure}_k = 1 - \text{mean}_i(V_{i,k} > 0)$$

Postsynaptic paired-pulse ratio

Used in: fig9

The EPSC-amplitude analogue of the presynaptic, Pr-based facilitation ratio, computed independently from the postsynaptic recording.

$$\text{PPR} = \text{Amplitude}_2 / \text{Amplitude}_1$$

6. Train reliability & variability metrics

fig10_sustained_unreliability (canonical); fig11_sync_async_reliability, figS4_paired_pulse_ISI_sync_coupling (near-identical copies)

Per-pulse coefficient of variation

Used in: fig10, fig11, figS4

$$\text{CV}_k = \text{SD}(\text{counts}_k) / \text{mean}(\text{counts}_k)$$

across trials, per pulse k ; also applied to VDCC influx (sampled 1 ms after each pulse's nominal onset) and to $[Ca^{2+}]_{AZ}$ over the train.

Bootstrap release probability

Used in: fig10, fig11, figS4

1000 with-replacement resamples of trials; the bootstrap distribution's mean and SD are reported as Pr_k and its CV.

$$Pr_k = \text{mean}(\text{success}_k) \text{ over resampled trials}$$

Consecutive-pulse conditional probability

Used in: fig10, fig11, figS4

The train-protocol analogue of the paired-pulse coupling metric P11.

$$P(k, k+1) = \# \{ \text{trials released at both } k \text{ and } k+1 \} / \# \{ \text{trials released at } k \}$$

Mutual information: pulse identity vs. release count

Used in: fig10 (also feeds fig10's summary-figure panel f)

Shuffle-corrected mutual information between which pulse a trial is on (X) and how many vesicles it released (Y), plus a per-pulse decomposition.

$$I(X;Y) = \sum_{x,y} p(x,y) \cdot \log_2 [p(x,y) / (p(x) \cdot p(y))]$$

$$I_{\text{corrected}} = I_{\text{raw}} - \text{mean}(I_{\text{raw}} \text{ over 200 label-shuffles })$$

$$\text{per-pulse contribution: } KL_x = [\sum_y p(x,y) \cdot \log_2 (p(x,y)/(p(x)p(y)))] / p(x)$$

X = pulse index, Y = release count per trial.

7. Synchronous / asynchronous release coupling

fig11_sync_async_reliability (canonical); figS4_paired_pulse_ISI_sync_coupling (paired with synchronous instead of asynchronous CV)

Sync / async fraction and CV

Used in: fig11, figS4

$$\text{async_fraction} = \text{mean}(\text{async}) / [\text{mean}(\text{sync}) + \text{mean}(\text{async})], \quad \text{sync_fraction} = 1 - \text{async_fraction}$$

$$CV_{\text{pop}} = \text{SD}(\text{pop counts}) / \text{mean}(\text{pop counts})$$

Pre-pulse residual active-zone calcium

Used in: fig11, figS4

The cross-trial-mean AZ Ca^{2+} level at the instant pulse k fires — calcium left over from prior pulses.

$$\text{residual}_k = \text{mean}_{\text{trials}} [[\text{Ca}^{2+}]_{\text{AZ}}(t) \text{ at } t = \text{onset of pulse } k]$$

Correlation and linear trend

Used in: fig11, figS4

Relates residual calcium to release-timing variability across pulses, within a condition and as an AD–Control difference.

$$r = \text{corrcoef}(\text{residual}, CV_{\text{async}}) \quad (\text{also computed on the AD–Control gap in each quantity, across pulses})$$

$$CV = m \cdot \text{residual} + c \quad (\text{least-squares line, overlaid on the scatter panel})$$